

Interrupt-Initiated I/O - MCQs and Time Delay Problems

1. What is the key difference between Programmed I/O and Interrupt-Initiated I/O?

- a) Interrupt-Initiated I/O requires continuous CPU involvement
- b) In Programmed I/O, the CPU waits for the device, whereas in Interrupt-Initiated I/O, the CPU executes other tasks until interrupted
- c) Programmed I/O is faster than Interrupt-Initiated I/O
- d) Interrupt-Initiated I/O does not require an Interrupt Service Routine (ISR)

✓ Answer: b) In Programmed I/O, the CPU waits for the device, whereas in Interrupt-Initiated I/O, the CPU executes other tasks until interrupted

2. Which technique helps the processor to run a program concurrently with I/O operations?

- a) Interrupt driven I/O
- b) DMA
- c) IOP
- d) DCP

✓ Answer: a) Interrupt driven I/O

3. In Interrupt-Initiated I/O, what is the role of the Interrupt Service Routine (ISR)?

- a) It tells the CPU to stop executing programs
- b) It services the I/O request and returns control to the main program
- c) It blocks further I/O operations
- d) It disables all interrupts

✓ Answer: b) It services the I/O request and returns control to the main program

4. What happens if a high-priority interrupt occurs while the CPU is servicing a lower-priority interrupt in a system with nested interrupts?

- a) The CPU ignores the high-priority interrupt
- b) The high-priority interrupt is serviced immediately, and the lower-priority ISR resumes later
- c) Both interrupts are serviced simultaneously
- d) The CPU crashes

✓ Answer: b) The high-priority interrupt is serviced immediately, and the lower-priority ISR resumes later

5. In a Daisy Chain Interrupt system, which of the following statements is true?

- A) All devices receive the interrupt acknowledge (INTA) signal simultaneously
- B) The closest device to the CPU has the lowest priority
- C) The highest-priority device receives the INTA signal first
- D) The CPU manually selects which device to service first

✓ Answer: C) The highest-priority device receives the INTA signal first

Explanation:

In a Daisy Chain system, the **INTA signal is passed down the chain**, with **higher-priority devices closer to the CPU** getting acknowledged first.

6. What is a major drawback of the daisy chain priority method?

- a) It does not support multiple devices
- b) Lower-priority devices may experience long delays
- c) It cannot handle interrupts efficiently
- d) It does not allow nested interrupts

✓ Answer: b) Lower-priority devices may experience long delays

7. In a parallel priority interrupt system, how is the highest-priority interrupt determined?

- a) By using a priority encoder
- b) By polling each device sequentially
- c) By giving equal priority to all devices
- d) By using a daisy chain

✓ Answer: a) By using a priority encoder

8. Which of the following best describes the sequence of operations in an interrupt-driven I/O system?

- a) Device sends an interrupt → CPU stops current task → CPU executes ISR → CPU resumes execution
- b) CPU checks device status → CPU processes request → CPU resumes execution
- c) CPU polls the device periodically → Device responds when ready
- d) CPU transfers data without device involvement

✓ Answer: a) Device sends an interrupt → CPU stops current task → CPU executes ISR → CPU resumes execution

9. A 50 kbps device is connected to a processor. The interrupt overhead is 50 μ s. The minimum performance achieved when an interrupt is initiated and data transferred is used instead of programmed I/O is:

- a) 2.4
- b) 0.4
- c) 3
- d) 3.5

✓ Answer: b) 0.4

Explanation:

Performance achieved when interrupt-driven used over programmed I/O:

$$S = ET_{\text{prog-IO}} / ET_{\text{INT-IO}}$$

$$S = 20 / 50 = 0.4$$

10. Which of the following statements about daisy chain interrupts is FALSE?

- A) It requires only a single interrupt request line
- B) The highest-priority device gets serviced first
- C) A failed device can block lower-priority devices
- D) The CPU services all devices in parallel

✓ Answer: D) The CPU services all devices in parallel

Explanation: In a daisy chain, **only one device is serviced at a time**, passing the interrupt acknowledge (INTA) signal down the chain.

11. A CPU with an interrupt-driven I/O system has a clock cycle of 1 ns. If the interrupt latency is 500 cycles and the ISR execution takes 250 cycles, how much time does the system spend handling each interrupt?

- a) 250 ns
- b) 500 ns
- c) 750 ns
- d) 1 μ s

✓ Answer: c) 750 ns

Explanation:

$$\begin{aligned} \text{Interrupt handling time} &= \text{Interrupt Latency} + \text{ISR Execution Time} \\ &= 500 \text{ cycles} \times 1 \text{ ns} + 250 \text{ cycles} \times 1 \text{ ns} \\ &= 750 \text{ ns} \end{aligned}$$

12. How can the processor ignore other interrupts when it is servicing one

- a) By turning off the interrupt request line
- b) By disabling the devices from sending the interrupts

- c) BY using edge-triggered request lines
- d) All of the mentioned

Answer: d

13. The return address from the interrupt-service routine is stored on the _____

- a) System heap
- b) Processor register
- c) Processor stack
- d) Memory

Answer: c

Explanation: The Processor after servicing the interrupts as to load the address of the previous process and this address is stored in the stack.

14. If an interrupt occurs while another interrupt is being serviced, what happens in a system with non-maskable interrupts (NMIs)?

- A) The new interrupt is ignored
- B) The current interrupt is paused, and the NMI is serviced immediately
- C) The CPU crashes
- D) Both interrupts are serviced simultaneously

✓ **Answer:** B) The current interrupt is paused, and the NMI is serviced immediately

Explanation:

NMIs are high-priority interrupts that cannot be disabled. The CPU **pauses the current ISR** and services the NMI immediately.

Time Delay Problems

15. A computer system has 4 devices connected in a Daisy Chain Priority Interrupt system. The CPU takes 2 μ s to respond to an interrupt request, and the propagation delay per device is 500 ns. If the fourth device in the chain raises an interrupt, how much time does it take before the CPU starts executing its Interrupt Service Routine (ISR)?

✓ **Solution:**

Total Propagation Delay = $(4 - 1) \times 500 \text{ ns} = 1500 \text{ ns} = 1.5 \mu\text{s}$
CPU Response Time = 2 μs

Total Delay Before ISR Execution = $2\ \mu\text{s} + 1.5\ \mu\text{s} = 3.5\ \mu\text{s}$

✓ Final Answer: $3.5\ \mu\text{s}$

16. A parallel priority interrupt system has 5 devices, each with a unique priority level. The CPU takes $3\ \mu\text{s}$ to respond to an interrupt and requires $2\ \mu\text{s}$ to resolve priority among multiple interrupting devices. If devices 2, 3, and 5 generate interrupts simultaneously, and device 3 has the highest priority, how much time elapses before the CPU begins executing the ISR for device 3?

✓ Solution:

Priority Resolution Time = $2\ \mu\text{s}$

CPU Response Time = $3\ \mu\text{s}$

Total Delay Before ISR Execution = $2\ \mu\text{s} + 3\ \mu\text{s} = 5\ \mu\text{s}$

✓ Final Answer: $5\ \mu\text{s}$